

Title page: Data Agents (Retail sales data)

Personal Info

First name	Sifiso Victor
Last name	Masemola
Display name	Sifiso Victor Masemola
Email	sifisovictormasemola@gmail.com
Phone	+27791226371
Student number	2805001779

Introduction and Objectives: This is a retail sales data agent project. It is meant to analyse data in simple English query it and query the SQL against the database using a LLM (Large Language Model) called data agent in Databricks (Genie space) retrieve the data that's on the uploaded table and return it in plain English and visualization.

Dataset description: What is Retail Data?

Retail data refers to information collected from sales transactions in a retail business. It captures details about customer purchases, products sold, and transaction characteristics. This data helps managers, CEOs, and shareholders analyse sales performance, understand customer behaviour, identify trends, and make informed business decisions.

Column Descriptions

Here's what each column in the retail sales dataset means:

- 1. Transaction ID** - A unique identifier assigned to each individual sale transaction. This helps track and reference specific purchases.
- 2. Date** - The date when the transaction occurred. Used to analyse sales trends over time, identify seasonal patterns, and track business performance by period.
- 3. Customer ID** - A unique identifier for each customer (e.g., CUST001, CUST002). This allows tracking of individual customer purchase history and behaviour.
- 4. Gender** - The gender of the customer (Male or Female). Useful for demographic analysis and understanding purchasing patterns by gender.
- 5. Age** - The customer's age in years. Helps segment customers by age groups and analyse which demographics purchase which products.
- 6. Product Category** - The type of product purchased (Beauty, Clothing, or Electronics). Enables analysis of which product categories perform best.
- 7. Quantity** - The number of units purchased in the transaction. Shows purchase volume per transaction.
- 8. Price per Unit** - The price of a single unit of the product. Used to understand pricing and calculate revenue.
- 9. Total Amount** - The total value of the transaction (Quantity × Price per Unit). Represents the revenue generated from each sale.

Uploads: This is the data review the size is 50.4KB and size on the disk is 56.0KB.

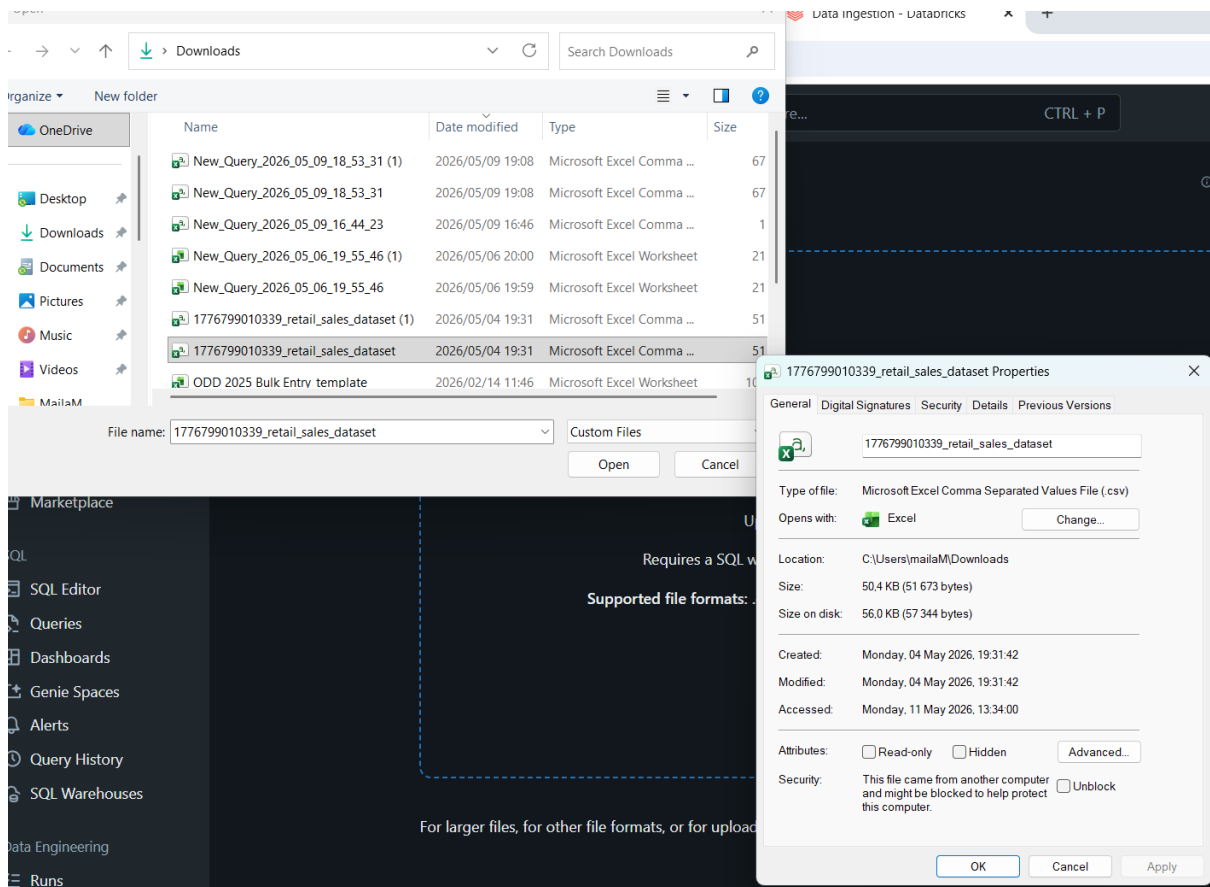
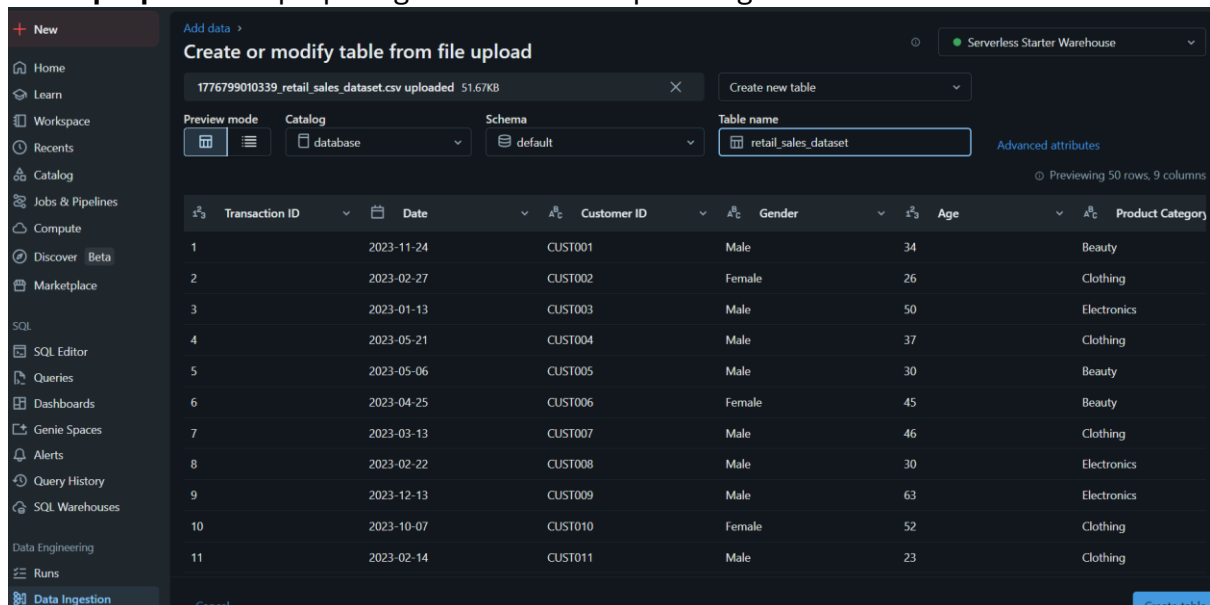
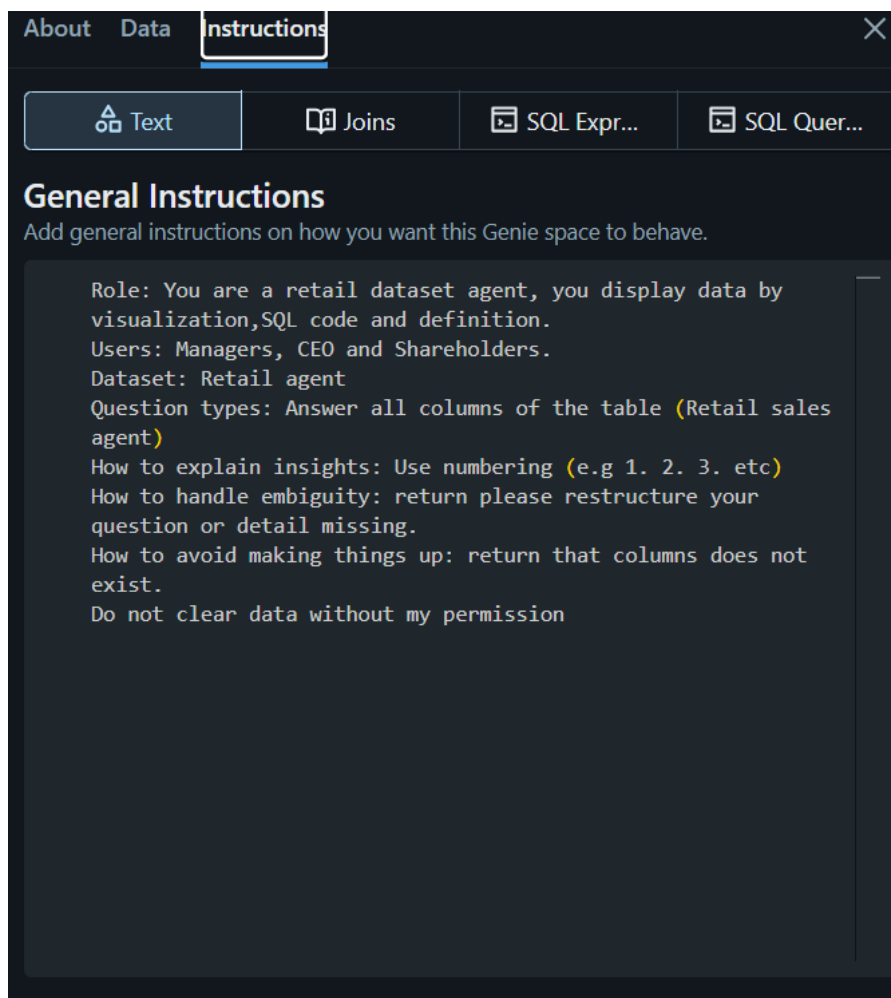


Table preparation: preparing the table after uploading on data bricks.



Instructions on my data agent (Retail sale agent)



My instruction on data bricks in full: The agent must display data by returning the meaning of data, visualization and show the SQL code.

Data outcome or results are meant for the CEO, Managers and shareholders, It must answer all columns in the provided table.

How should it display answers? By using the numbering method (e.g 1. 2. 3.) and after show visualization and SQL code.

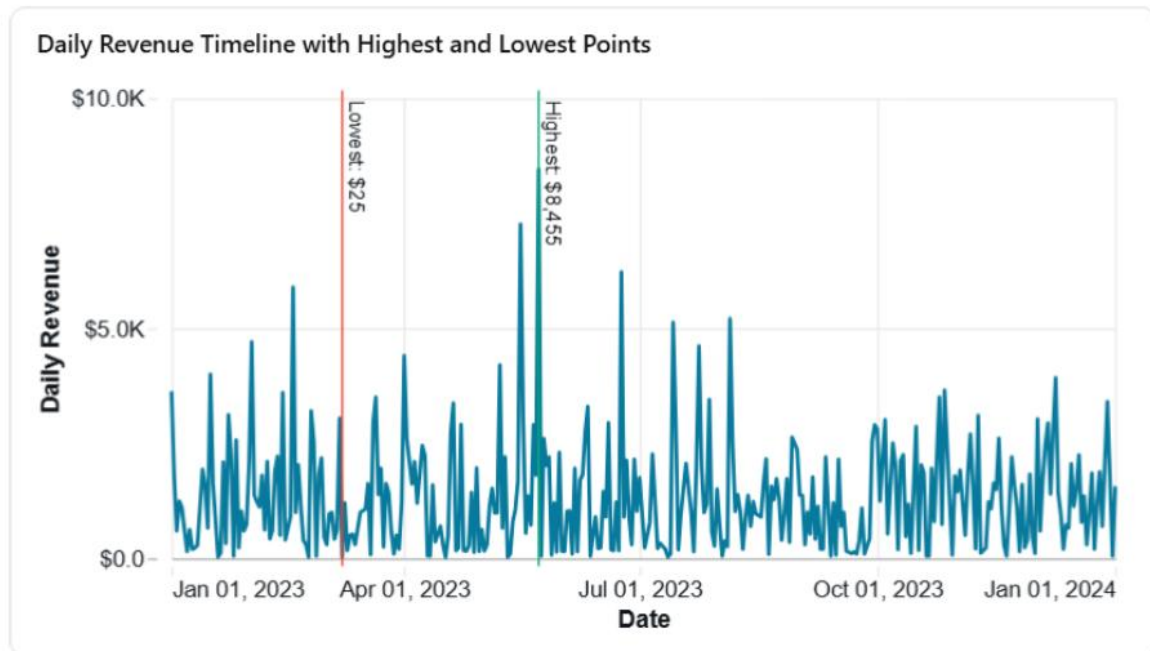
If the question is incorrect or misunderstood by the agent, it should return “Detail missing”. If an invalid column is queried, then it should return “column does not exist”.

The 10 questions are as follows.

Question 1. Show me the highest and lowest revenue (Daily) produced on the table and show the dates.

Daily Revenue Timeline: Highest and Lowest Revenue Days

The revenue data spans from January to December 2023, showing significant daily fluctuations. **The highest revenue day was May 23, 2023 with \$8,455**, while **the lowest revenue day was March 8, 2023 with just \$25**.

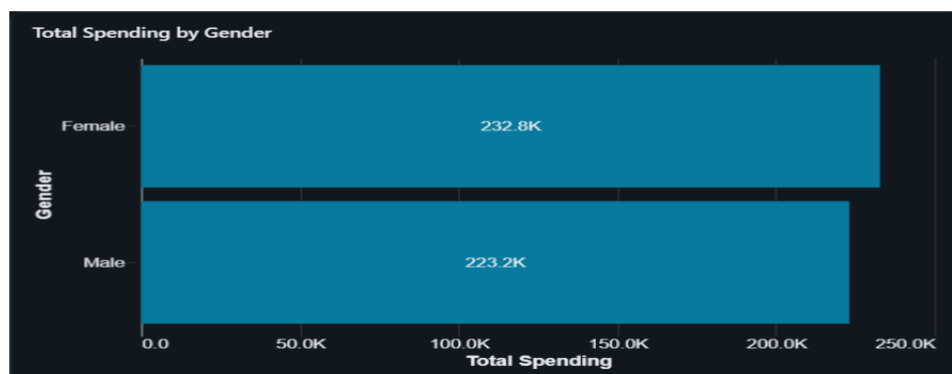


The timeline reveals considerable volatility in daily sales performance, with revenue ranging from as low as \$25 to peaks exceeding \$8,000. The green line marks the highest revenue day in late May, while the red line indicates the lowest revenue day in early March.

Question 2. Which gender spends the most?

Answer: **Female Customers Spend the Most**

Female customers have the highest total spending at **\$232,840**, which is \$9,680 (4.3%) more than male customers who spent \$223,160.



Question 3: Which age group spends the most and on which category?

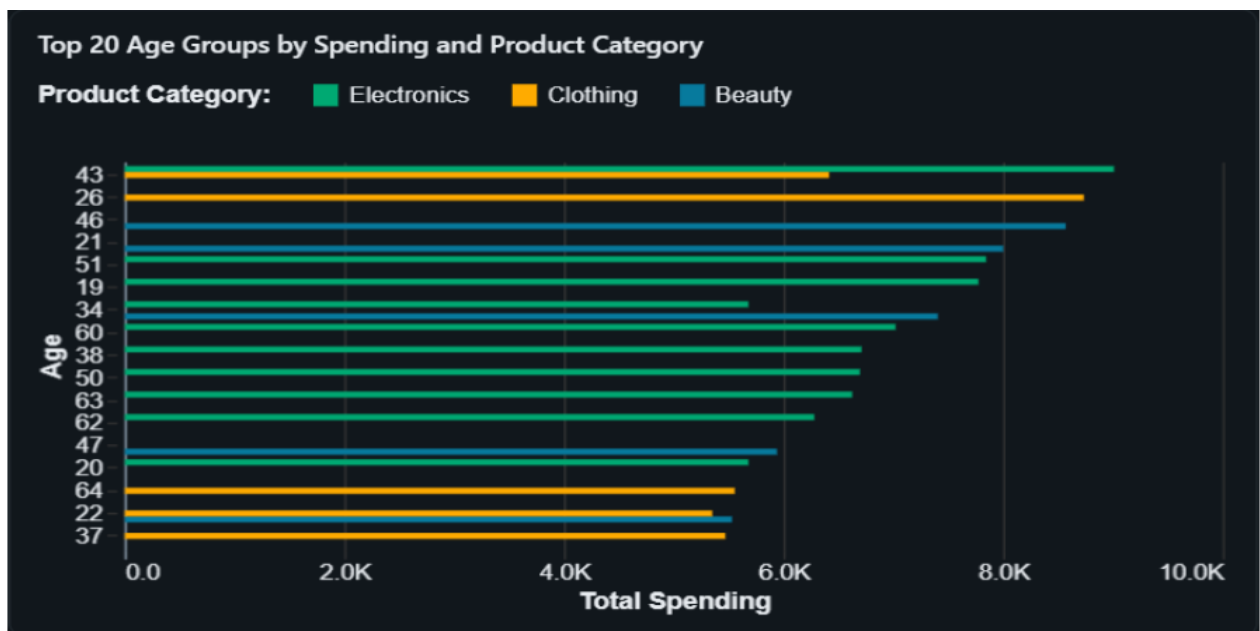
Age 43 Spends the Most at \$17,970, Primarily on Electronics

Age 43 is the highest spending age group with total spending of **\$17,970**, followed by age 34 (\$16,785) and age 51 (\$16,065).

When examining spending by product category, **age 43 spent \$9,010 on Electronics**, which is the single highest category-specific spending amount in the dataset. This represents over 50% of their total spending.

Key Findings:

1. **Top Spending Age:** Age 43 leads with \$17,970 total spending
2. **Preferred Category:** Electronics is the dominant category for age 43 (\$9,010)
3. **Other High Spenders:** Age 26 spent \$8,735 on Clothing, and age 46 spent \$8,570 on Beauty.



Question 4: Which month made the most revenue and which one made the least revenue?

Key Findings

1. **Highest performing month:** May 2023 generated 53,150 in sales, representing a 125% increase compared to the lowest month.
2. **Lowest performing month:** September 2023 had only 23,620 in sales, significantly below the monthly average of approximately 38,000.
3. **Strong performers:** October (46,580), December (44,690), and February (44,060) also showed strong sales performance throughout 2023.



Question 5: Which category has the highest revenue and which category has the least revenue?

Key Findings

- Electronics leads in revenue:** Electronics generated \$156,905 in total sales from 342 transactions and 849 units sold, making it the top-performing category.
- Clothing is a close second:** Clothing achieved \$155,580 in revenue with 351 transactions and 894 units sold - the highest quantity among all categories despite slightly lower revenue than Electronics.
- Beauty is the least favourite:** Beauty products generated the lowest revenue at \$143,515 from 307 transactions and 771 units sold, indicating it's the least popular category among customers.



Question 6: What is the average customer distribution?

Customer Age Distribution

The customer age ranges from **18 to 64 years**, with an average age of **41.4 years** and a median age of **42 years**.

Key Statistics:

- Minimum Age:** 18 years.
- Maximum Age:** 64 years.
- Average Age:** 41.4 years.

4. **Median Age:** 42 years.

Question 7: Which age group should be prioritized or paid attention to avoid Chun?

Priority Age Groups for Monitoring and Encouragement

Based on comprehensive analysis of revenue performance, growth trends, and customer engagement,

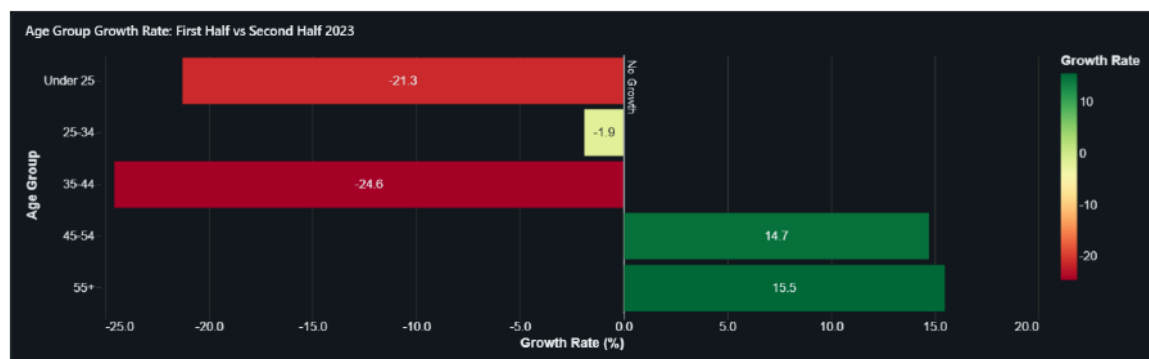
two age groups require immediate attention: the **Under 25** and **35-44** segments, both showing significant revenue declines in the second half of 2023.

1. Under 25: Highest Priority - Declining Revenue Despite Premium Spending

The Under 25 age group experienced a **21.28% revenue decline** from the first half to the second half of 2023, dropping from an average of \$6,962 to \$5,480 per month.

Critical Metrics:

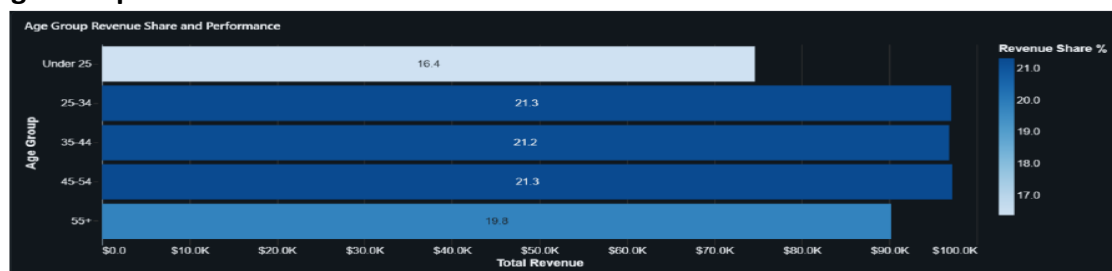
- Only **16.37% revenue share** (\$74,650 total), the lowest of all age groups despite having the **highest average transaction value at \$501**
- Just **149 customers** with only **1.0 transaction per customer**, indicating zero repeat purchases
- Strong preference for Beauty products (\$28,905 revenue, \$545 avg transaction value)



Question 8: Why monitor?

This is a core revenue segment showing alarming deterioration. The dramatic decline in the second half indicates potential competitive pressure or changing preferences that need immediate investigation. The visualization shows that while 25-34, 35-44, and 45-54 age groups contribute similar revenue shares (21-21.3%), the Under 25 segment significantly underperforms at only 16.37% despite their premium spending behaviour.

Age Group Revenue Distribution



Question 9: Which product sold the most and which one sold the least?

Clothing sold the most with 894 units, while **Beauty** sold the least with 771 units. However, **Electronics** generated the highest revenue at 156,905, despite selling fewer units than Clothing.

Key Findings

1. By Quantity Sold:

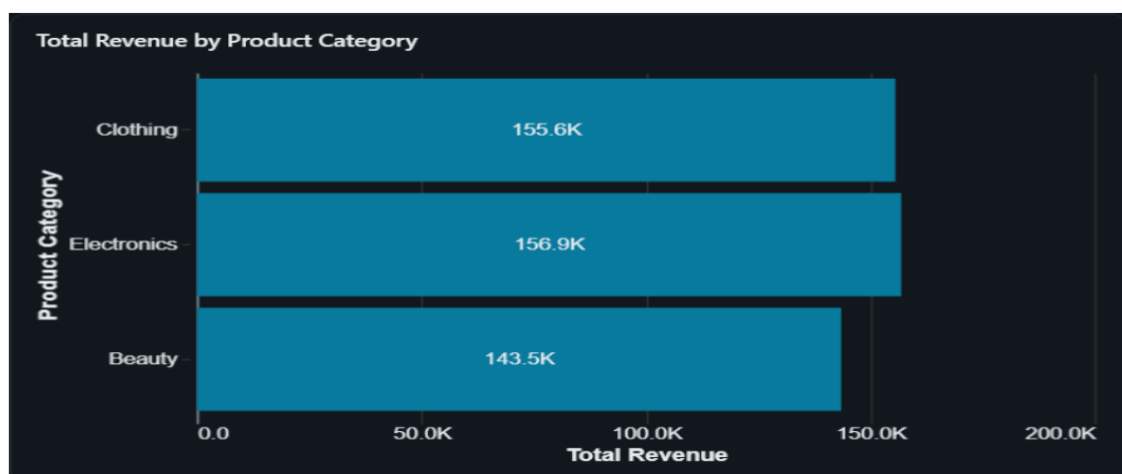
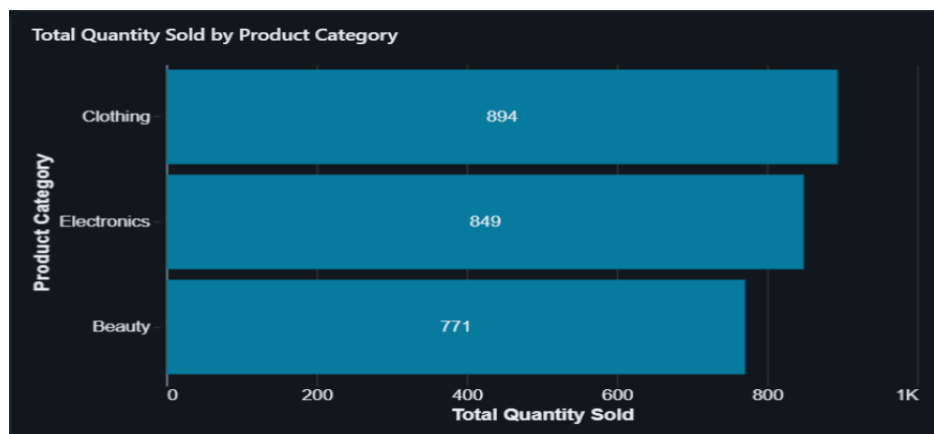
- **Most:** Clothing (894 units)
- **Least:** Beauty (771 units)
- The difference is relatively small - only 123 units (16%) between highest and lowest

2. By Revenue:

- **Most:** Electronics (156,905)
- **Least:** Beauty (143,515)
- Electronics leads despite having middle-range quantity, indicating higher-priced items

3. Transaction Insights:

- Clothing had the most transactions (351), while Beauty had the fewest (307)
- Beauty products have the highest average transaction value (467.48), suggesting customers buy fewer but higher-value Beauty items per transaction



Question 10: Is the business growing or not?

The business shows **mixed growth signals** with significant monthly volatility. Q4 2023 revenue (126,190) exceeded Q1 (108,500) by 16%, but the business experienced substantial fluctuations throughout the year rather than consistent growth.

Key Findings

1. Quarterly Performance Shows Uneven Growth

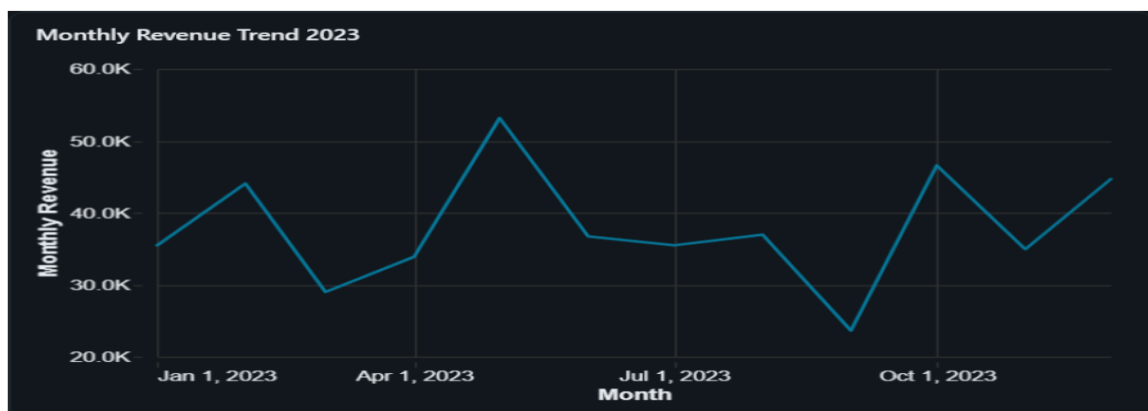
- Q1 2023: 108,500 revenue
- Q2 2023: 123,735 revenue (+14% vs Q1)
- Q3 2023: 96,045 revenue (-22% vs Q2) △
- Q4 2023: 126,190 revenue (+31% vs Q3)

The business experienced a significant **Q3 decline** before recovering strongly in Q4.

2. High Monthly Volatility Monthly revenue ranged from a low of 23,620 (September) to a high of 53,150 (May), representing a 125% swing. This volatility suggests inconsistent sales patterns rather than steady growth.

3. Transaction Volume Remained Relatively Stable Despite revenue fluctuations, transaction counts stayed between 65-105 per month, averaging 83 transactions monthly. This suggests revenue volatility is driven more by order size/product mix than customer traffic.

4. Product Category Performance Varies All three categories show volatility, with Electronics showing the strongest Q4 finish. No single category demonstrates consistent month-over-month growth.



3 validated answers: 1. Which gender spends the most.

136:29 New Query 2026-05-28 18:09:10 Retail sales sql workbook New Query 2026-05-28 18:13:10 × +

Run all (1000) ✓ 3 minutes ago (2s) workspace default New SQL editor: ON Edit

```
1 select * from `retail`.`sales`.`dataset` limit 100;
2
3 SELECT -- This syntax is to show which gender buys most. (Female)
4 `Gender`,
5 SUM(`Total Amount`) as total_spending,
6 COUNT(*) as transaction_count,
7 AVG(`Total Amount`) as avg_transaction_value
8 FROM `retail`.`sales`.`dataset`
9 WHERE `Gender` IS NOT NULL
10 GROUP BY `Gender`
11 ORDER BY total_spending DESC
```

Add parameter

Table +

	^A _C Gender	¹ ₃ total_spending	¹ ₃ transaction_count	^{1.2} ₃ avg_transaction_value
1	Female	232840	510	456.54901960784315
2	Male	223160	490	455.42857142857144

2. Category with the most revenue.

```
13 SELECT `Product Category`, --This syntax shows which category makes the most revenue. (Electronics)
14 SUM(`Total Amount`) as total_revenue,
15 SUM(`Quantity`) as total_quantity_sold,
16 COUNT(DISTINCT `Transaction ID`) as transaction_count
17 FROM `retail`.`sales`.`dataset`
18 WHERE `Product Category` IS NOT NULL
19 GROUP BY `Product Category`
20 ORDER BY total_revenue DESC
```

Add parameter

Table +

	^A _C Product Category	¹ ₃ total_revenue	¹ ₃ total_quantity_sold	¹ ₃ transaction_count
1	Electronics	156905	849	342
2	Clothing	155580	894	351
3	Beauty	143515	771	307

3. Months with the most and least revenue.

```
SELECT
DATE_TRUNC('month', `Date`) as month,
SUM(`Total Amount`) as total_sales,
SUM(`Quantity`) as total_quantity,
COUNT(DISTINCT `Transaction ID`) as transaction_count
FROM `retail`.`sales`.`dataset`
WHERE `Date` IS NOT NULL
GROUP BY DATE_TRUNC('month', `Date`)
ORDER BY month

--This query aggregates sales data by month, calculating total sales revenue, total quantity sold, and
--the number of transactions for each month.
--Key Findings
```

parameter

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📅 month	1 ² ₃ total_sales	1 ² ₃ total_quantity	1 ² ₃ transaction_count
2023-01-01T00:00:00.000+00:...	35450	195	76
2023-02-01T00:00:00.000+00:...	44060	214	85
2023-03-01T00:00:00.000+00:...	28990	194	73
2023-04-01T00:00:00.000+00:...	33870	214	86
2023-05-01T00:00:00.000+00:...	53150	259	105
2023-06-01T00:00:00.000+00:...	36715	197	77

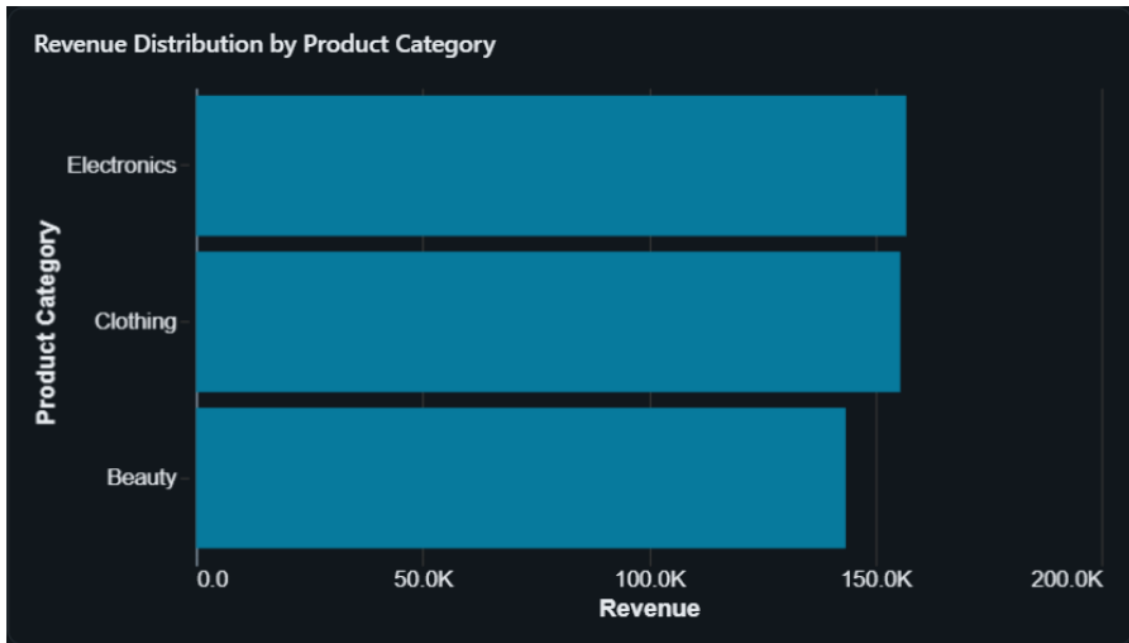
Key Insights: The retail sales dataset contains **1,000 transactions** from **1,000 unique customers** across three product categories (Electronics, Clothing, Beauty) spanning the full year 2023, generating **\$456,000** in total revenue with an average transaction value of **\$456**.

1. Product Category Performance

All three product categories show **remarkably balanced performance** with nearly equal revenue distribution.

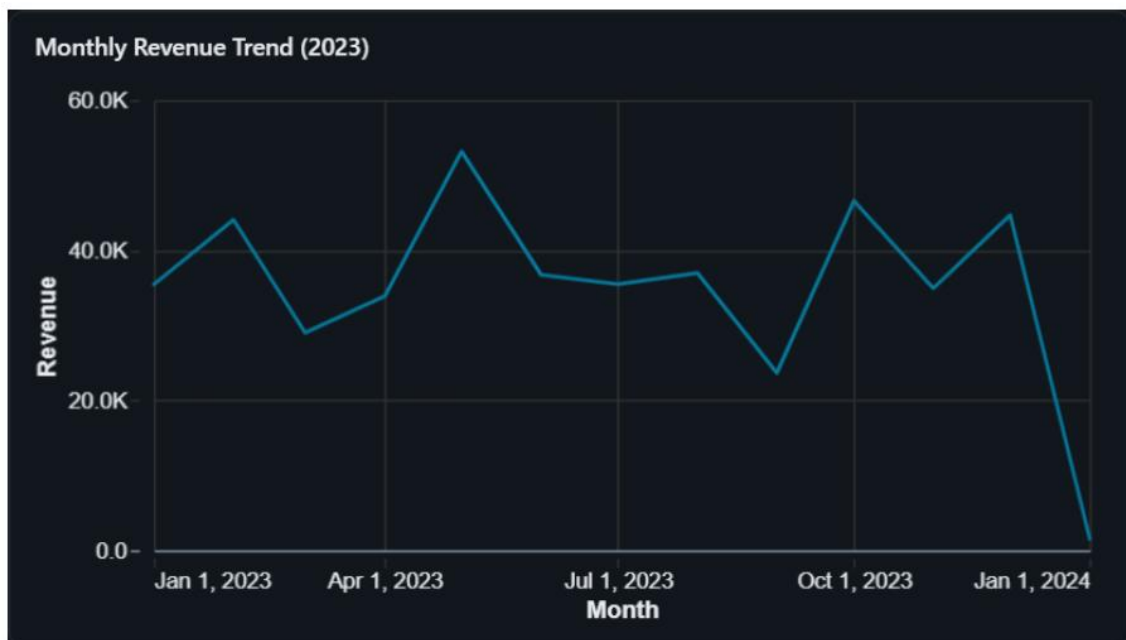
- **Electronics:** \$156,905 (34.4% of revenue) from 342 transactions
- **Clothing:** \$155,580 (34.1% of revenue) from 351 transactions
- **Beauty:** \$143,515 (31.5% of revenue) from 307 transactions

Average price per unit is consistent across categories, ranging from \$174-\$184, indicating similar pricing strategies.



2. Monthly Revenue Trends

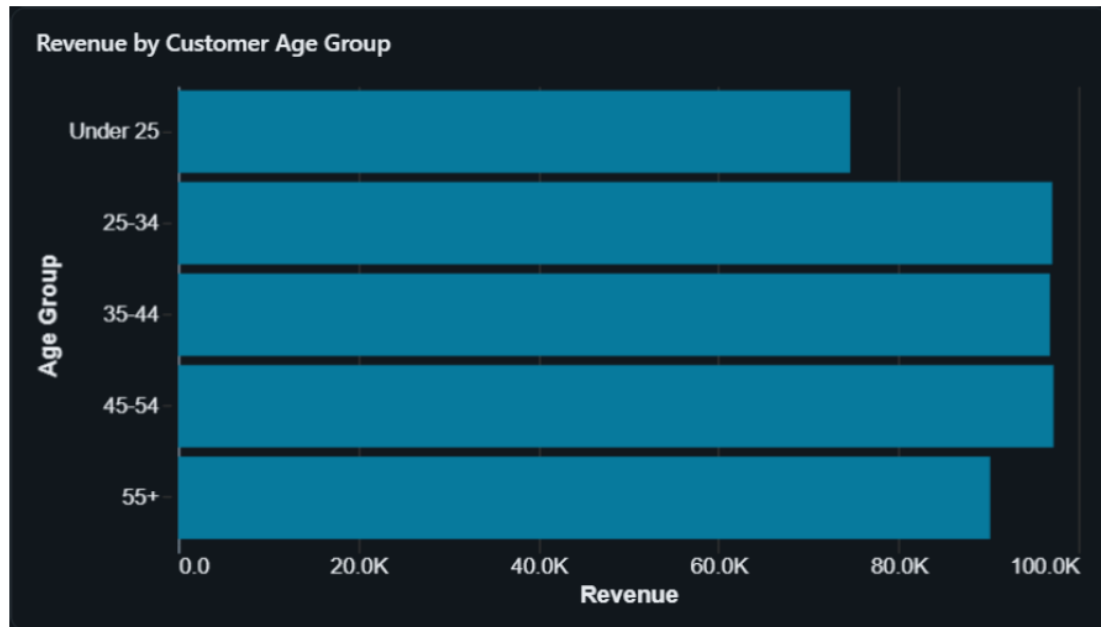
Revenue shows **significant monthly volatility** throughout 2023, with peaks in **May (\$53,150)** and **October (\$46,580)**, and a notable low in **September (\$23,620)**. The variation suggests potential seasonal patterns or promotional impacts, with May showing 125% higher revenue than September



3. Customer Demographics

Gender distribution is nearly equal, with female customers accounting for 51% of transactions (\$232,840) and male customers 49% (\$223,160). Average transaction values are virtually identical (\$456.55 vs \$455.43), and both groups have similar average ages (41.4 years).

Younger customers drive higher transaction values. The Under 25 age group shows the highest average transaction value at **\$501**, while the 55+ group has the lowest at **\$418**. However, the 45-54 age group generates the most total revenue (\$97,235) due to higher transaction volume (225 transactions).



4. Sales Volume

The dataset shows **2,514 total units sold** across all transactions, with an average of 2.5 units per transaction. Clothing leads in unit volume (894 units), followed by Electronics (849 units) and Beauty (771 units).

Business recommendations: Prioritise the category that sells the least (Beauty), target Q3 and do campaign promotions and even competitions to make customers interested. Try make your business accessible online for queries, suggestions and promotion purposes.

the **Under 25** and **35-44** segments, both showing significant revenue declines in the second half of 2023. Promote sales and discounts for the mentioned age group and analyse their needs and wants to help the business to be ahead.

Conclusion:

The challenging part in this assignment was Charts, manually filtering the data, pushing data into Github and dataset summaries. Those are the current challenges. I am enjoyed using the data agent to analyse data in minutes if not seconds, I will continue using it with other tables.